

Curriculum Based Measurement in Math Calculations

Purpose, use, and administration guidelines of CBM-M

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Overview of CBM-Math

- Curriculum Based Measurement aims to index the level and rate of student achievement in math.
- Monitor intervention effectiveness
- Data-Based Decision Making[3,4]
  - ***Screening***
    - Local grade level norms at the classroom, school, and district level
  - ***Benchmarking***
  - ***Progress monitoring***
    - Establishes baseline
    - Standardized procedures to measure growth overtime

CBM-M Research

- CBM-M are relevant and useful in making decisions[2]:
  - Research is specific to CBM-Math computations and is useful and appropriate for making instructional decisions based on screening applications.
  - Assessments that target computation skills exclusively have acceptable reliability and validity.
  - Useful in planning, tailoring, and adapting instruction to the needs of students.



Overview of CBM-M Procedures

- Students write answers to mathematical computation problems.
- Administrators can choose between several types of probes, including
  1. Math Fact, Single or Multiple Skill
  2. Grade-level Probes
- Probes consist of 2 pages of problems.



Overview of CBM-M Procedures: Following Standardized Procedures

- Must give the test the same way with each administration.
- Standardized procedures:
  - Directions
  - Scoring procedures
  - Probe type



Overview of CBM-M Procedures: Math Fact Probe Types

- Addition Fact Families
- Subtraction Fact Families
- Addition and Subtraction Fact Families
- Multiplication Fact Families
- Division Fact Families
- Multiplication and Division Fact Families
- Addition, Subtraction, Multiplication, and Division Fact Families (Multi-Skill Probe)



Overview of CBM-M: Grade-Level (Multi-skill) Probes

- Where single skill probes include problems of one type, multi-skill probes include the same proportion of problem type

Example

Grade One Computational Problems

Subjects	Proportion
Addition facts sums to 18	30%
Subtraction facts from 9	30%
Addition sums to 35	20%
Subtraction from 20	20%



Administering CBM-M

- In order for CBM-M administration to be reliable and accurate, it is important to make sure that:
  - All directions are followed exactly.
  - All errors are marked appropriately.
  - The last problem completed is marked appropriately.
  - No errors are corrected.
  - No feedback is given.



Administering CBM-M

Materials

- CBM-M worksheet
  - Single skill
  - Multiple skill
- Examiner sheet (key)
- Timer
- Pen/pencil student

Monitor students

- Discourage erasing
- Keep on task



Administering CBM-M: Directions

Directions

- "We're going to take a _X_ minute Math test. When I say 'BEGIN,' write your answer to the FIRST problem and work ACROSS the page (point to demonstrate). Then go to the next row. Try to EACH problem. If you come to one YOU REALLY DON'T KNOW HOW TO DO, put an 'X' through it and go to the next one. If you finish the first page the go to the next page and continue working. Continue working until I say, 'Stop.' Are there any questions? (Pause). Begin"

- "Stop and put your pencils down."



Administering CBM-M: Time Recommendations

- Time allotted for administration depends on the probe type:
  - Facts and Single-skill
    - 2 minutes
  - Basic Multi-skill
    - 4 minutes
  - Complex (Carrying, borrowing, long division, etc.)
    - 6 minutes



Scoring CBM-M

- Several types of scoring procedures exist:
  - Digits Correct (DC; most sensitive)
    - Digits within the answer only
    - Digits within the critical process and answer
      - e.g., long division
  - Number of problems correct
  - Partial credit
- It is most important to be consistent in applying the same scoring procedure across administrations.



Scoring CBM-M: General Rules

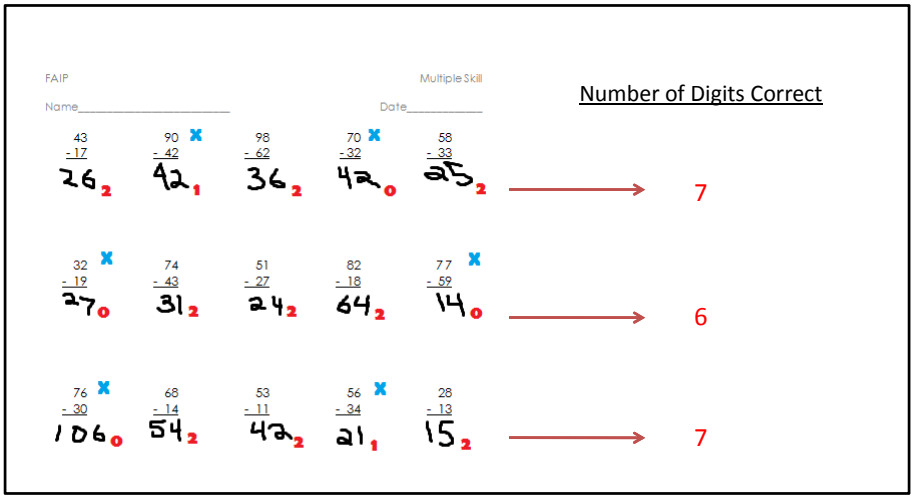
- X-ed Out Problems
  - If a student writes an answer for a problem he/she has crossed out, ignore the X and score what you see.
- Incomplete Problems: if problem is not finished, score for number of DC's written.
- Legibility/Reversed/Rotated Numbers:
  1. Cannot interpret, then count it wrong.
  2. Correct with reversed number is scored as correct.

Use your best judgment

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Scoring CBM-M: Digits Correct



Total DC = 22

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Scoring CBM-M: Digits Correct

- Digits Correct using critical processes with answer

Correct Answer	Wrong Answer with No Work	Answer Key
502 x 61 = 502, 3012, 30622 Possible = 12 DC Score = 12 DC	502 x 61 = 563 Possible = 12 DC Score = 0 DC	502 x 61 = 502, 3012, 30622

Student earns 12 digits correct for showing work with correct digits, and for the correct answer.

Student earns zero digits correct for not showing work and incorrect answer



Scoring CBM-M: Answer & Critical Processes

- **Alignment:**
  - If answer is correct, ignore alignment problem and count digits correct.
  - If incorrect, count digits as they appear in approximate place value, even if place value error seems obvious.

Student earns three digits correct

Incorrect Alignment	Answer Key
502 x 61 = 502, 3012 Possible = 12 CD Score = 3 CD	502 x 61 = 502, 3012, 30622



Scoring CBM-M: Answer & Critical Processes

- Some students don't show all their work
  - If answer written is **correct, their score is the number of DC's possible.**
  - If answer written is **incorrect, examiner can only score what they see.**

***To do this: underline and count any digits that are correct, even if the answer is incorrect.**



Example Answer Sheet

FAIP Multiple Skill

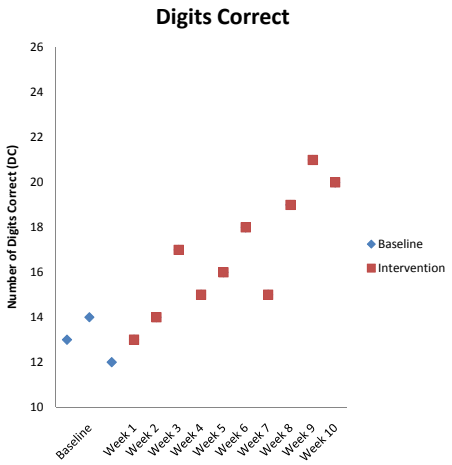
Name_____ Date_____

76 + 67 = 143 (3)	14 + 50 = 64 (2)	86 + 83 = 169 (3)	95 + 45 = 140 (3)	10 + 74 = 84 (2)
55 + 82 = 137 (3)	90 + 57 = 147 (3)	41 + 91 = 132 (3)	69 + 46 = 115 (3)	11 + 92 = 103 (3)
49 + 48 = 97 (2)	70 + 63 = 133 (3)	98 + 25 = 123 (3)	53 + 61 = 114 (3)	31 + 88 = 119 (3)
39 + 21 = 60 (2)	16 + 88 = 104 (3)	37 + 79 = 116 (3)	35 + 87 = 122 (3)	78 + 71 = 149 (3)



Graphing CBM-M

1. Establish baseline
2. Set student goal
3. Progress monitor

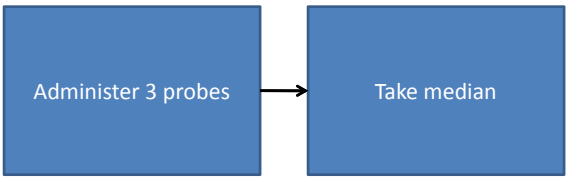


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Graphing CBM-M: Procedures

1. Establish Baseline



Digits Correct

Day 1	Day 2	Day 3
13	14	12

Baseline 13

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Graphing CBM-R: Procedures

2. Goal should be ambitious, yet attainable.

3. Establish Progress Monitoring Schedule
   - Define number of weeks

Rates of weekly increase for Digits Correct[1]	
Grade 1	0.35
Grade 2	0.30
Grade 3	0.30
Grade 4	0.70
Grade 5	0.70
Grade 6	0.40

Goal = (Expected Rate of Growth x Weeks) + Baseline

- Baseline = 13 DC
- Growth Rate = .70
- Weeks = 10

(.7 × 10) + 13
Goal = 20 Digits Correct

[1]National Center for Progress Monitoring

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Digits Correct Standards

Grade	Level	Median Digits Correct per Minute	Median Digits Incorrect per Minute[6]
Grades 1-3	Frustrational Instructional Mastery	0-9 10-19 20+	8+ 3-7 ≥2
Grade 4+	Frustrational Instructional Mastery	0-19 20-39 40+	8+ 3-7 ≥2

- Fluency data are more reliable than accuracy data.
- Instructional level:[1]
  - 14-31 digits correct per minute for 2nd and 3rd grade.
  - 24-49 digits correct per minute for 4th and 5th grade.
  - Districts may also have their own standards.

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Summary

- This module provides an overview of CBM-M use, procedures, and scoring.
- Encourage students to continue working, to try each problem and not to skip problems.
- Digits are scored for their correct placement by place value.
- All digits correct are earned for correct answers.



Resources

Intervention Central

- http://www.Interventioncentral.org
  - Generate CBM probes
  - EB Intervention ideas
  - And more!

AIMS Web

- http://www.aimsweb.com (subscription required)
  - Popular CBM probes
  - Training modules
  - Downloads



References

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2. Christ, T. J., Scullin, S., Tolbize, A., & Jiban, C. L. (2008). Implications of Recent Research: Curriculum-Based Measurement of Math Computation. *Assessment for Effective Intervention, 33*(4), 198-205.
3. Deno, S. L. (1986). Formative evaluation of individual student programs: A new role for school psychologists. *School Psychology Review, 15*(3), 358-374.
4. Deno, S. L. (2003). Developments in Curriculum-Based Measurement. *Journal of Special Education, 37*(3), 184-192.
5. Fuchs, L. S., & Fuchs, D. (1986). Effects of systematic formative evaluation: A meta-analysis. *Exceptional Children, 53*, 3-14.
6. Shapiro, E. S. (2004). *Academic skills problems: Direct assessment and intervention* (3rd ed.). New York: Guildford Press.

Additional Training Modules

- Using CBM for Progress Monitoring in Reading (National Center on RTI)
- Classroom Assessment (Part 2): Evaluating Reading Progress: (Peabody Iris Center)
- Using CBM for Progress Monitoring in Reading (National Center on Student Progress Monitoring